

Lab stock + PDB purchasing brief

Purpose

Prepare a shared lab stock purchase for the new pick-and-place machine, plus the exact parts needed for PDB Rev A prototypes. Prepared September 14, 2026 for the September 15 discussion. Build quantity, machine model, budget and existing usable stock are not yet known.

Recommendation

Inventory Ryan's R/C/L books first. Request quotes for small machine-ready reels of the most reusable passives and compare with one factory reel per selected stock item. Buy project-specific ICs, MOSFETs, shunts and connectors in prototype quantities. Do not buy full semiconductor reels just because the order code is a reel variant.

What is confirmed

The A-P1 PDB baseline contains 27 unique purchased part codes, 33 physical items per board and 29 SMT placements across 23 SMT part types. Seven bare test pads do not require purchased parts. This list supports procurement planning; final circuit qualification and PCB layout remain open.

Osiris Rev B boundary

The new Rev B power path and top-facing ports are planned, not saved as a completed design. Its new regulator, inductor and power-path BOM is unknown. Historical Osiris passives are largely 0402, while this PDB uses 0603/0805/1206. Equal values in different packages cannot share a reel. No new Rev B quantities are included here.

Three purchasing buckets

A: reusable passive stock candidates, pending cross-project demand. B: less-used or specification-sensitive passives in small machine-ready quantities. C: specialized components and manual-assembly parts for the selected prototype batch. HOLD: provisional fuse rating.

What to ask for tomorrow

A budget for shared reel stock and a separate prototype-parts allowance, plus machine feeder details and an inventory of Ryan's books. The following pages provide every exact PDB order code without assuming a build count.

1 / Passive stock candidates

All package sizes below are EIA. A-marked parts are candidates for lab stock, not confirmed common parts on the redesigned Osiris Rev B. Quote 250 / 1,000 custom-reel quantities versus one factory reel.

Exact MPN / specification	Refs	Qty / PDB	Package	Buying class
C0603C104K4RACTU 100 nF, 16 V, X7R, 10%	C3, C5, C8	3	0603 (EIA)	A - stock candidate
C1608X7R1H104K080AA 100 nF, 50 V, X7R, 10%	C7	1	0603 (EIA)	A - stock candidate
C2012C0G2A562J125AA 5.6 nF, 100 V, C0G, 5%	C1	1	0805 (EIA)	B - small tape/reel
C2012X7R1A106K125AC 10 uF, 10 V, X7R, 10%	C6	1	0805 (EIA)	B - small tape/reel
C2012X7R1H105K125AB 1 uF, 50 V, X7R, 10%	C2	1	0805 (EIA)	B - small tape/reel
C3216X7R1H106K160AC 10 uF, 50 V, X7R, 10%	C4, C9	2	1206 (EIA)	B - small tape/reel
RC0805FR-0710RL 10 ohm, 1%, 0.125 W	R8, R9	2	0805 (EIA)	A - stock candidate
RC0805FR-07200KL 200k ohm, 1%, 0.125 W	R7	1	0805 (EIA)	B - small tape/reel
RC0805FR-0722KL 22k ohm, 1%, 0.125 W	R4	1	0805 (EIA)	A - stock candidate
RC0805FR-072ML 2M0 ohm, 1%, 0.125 W	R1	1	0805 (EIA)	B - small tape/reel
RC0805FR-0743K2L 43k2 ohm, 1%, 0.125 W	R2	1	0805 (EIA)	B - small tape/reel
RC0805FR-074K7L 4k7 ohm, 1%, 0.125 W	R10	1	0805 (EIA)	A - stock candidate
RC0805FR-0756K2L 56k2 ohm, 1%, 0.125 W	R3	1	0805 (EIA)	B - small tape/reel
RC0805FR-07680KL 680k ohm, 1%, 0.125 W	R6	1	0805 (EIA)	B - small tape/reel

B-marked parts: purchase to the prototype batch formula. Do not replace precision divider values, timing dielectric, voltage ratings or package sizes to match an assortment book. Confirm complete manufacturer specifications at quotation, including the 2 M and 680 k resistor order codes.

2 / Specialized PDB components

Purchase in small quantities based on the agreed build count. For SMT, ask for continuous tape or a custom reel compatible with the machine. Through-hole connectors and holder require a separate assembly operation.

Exact MPN / specification	Refs	Qty / PDB	Package	Buying class
0297004.WXNV 4A MINI	F1	1	MINI blade fuse	HOLD rating
74AUP1G07GW,125 74AUP1G07GW	U4	1	SOT-353	C - prototype SMT
BM04B-GHS-TBT(LF)(SN) OSIRIS_I2C	J2	1	JST GH 4-pin vertical	C - prototype SMT
FC4L64R005FER 5 milliohm, 4-terminal shunt	R5	1	6432 4-terminal	C - prototype SMT
INA228AIDGSR INA228AIDGSR	U2	1	VSSOP-10	C - prototype SMT
ISC015N06NM5LF2ATMA1 ISC015N06NM5LF2	Q1, Q2	2	TDSON-8 5x6 mm	C - prototype SMT
LTC4368IMS-1#TRPBF LTC4368IMS-1-PBF	U1	1	MSOP-10	C - prototype SMT
MBR0540T1G MBR0540	D2, D3	2	SOD-123	C - prototype SMT
SMAJ24CA SMAJ24CA	D1	1	SMA	C - prototype SMT
TPS7A1633DGNR TPS7A1633DGNR	U3	1	MSOP-8 exposed pad	C - prototype SMT
3568 MINI fuse holder	F1-holder	1	MINI fuse holder TH	C - manual/TH
XT60PW-F XT60PW-F	J3	1	XT60 horizontal TH	C - manual/TH
XT60PW-M XT60PW-M	J1	1	XT60 horizontal TH	C - manual/TH

The fuse holder is a separate purchased item. The 4 A fuse is a provisional selection: hold the final rating until the new Osiris load budget is approved. Protect the LTC4368 -1 variant, exact packages and specified manufacturer identities when quoting.

3 / Turn the list into an order

Ryan's books: inventory before buying

Record value, EIA package, manufacturer/MPN, tolerance, power or voltage rating, dielectric, quantity and carrier condition. Distinguish loose parts, short tape strips and continuous tape. Count stock toward machine assembly only after feeder compatibility is established. Unidentified parts are not approved substitutions for the PDB.

Shared stock proposal

For each of the five A-marked candidates, quote 250 and 1,000 parts on a custom reel, plus the smallest factory reel. These are quote tiers, not committed order quantities. Select one tier after checking other project BOMs, Ryan's stock and total cost including reeling. The two 100 nF parts remain separate MPNs; combining them needs a reviewed substitution.

Project-specific quantity rule

For each B or C item, order $\max(0, \text{boards to build} \times \text{quantity per board} + \text{setup losses} + \text{repair spares} - \text{usable on-hand stock})$, rounded to supplier pack quantity. Establish setup loss with the machine supplier; do not assume a percentage covers feeder loading. Request continuous cut tape or a custom reel for SMT parts; loose pieces are for supported trays/manual work only.

Machine and assembly purchases

Confirm machine model, 0402 capability, feeder widths and pitch, reel diameter, component height limits, nozzle selection, vision support and cut-tape/tray handling. There are 23 SMT part types on the PDB; that is not a guaranteed 23-slot feeder requirement because widths and staging vary. Budget feeders, suitable nozzles, leader/trailer supplies, ESD storage and moisture-control supplies as required by component handling specifications.

Items outside the PCB BOM

Budget solder paste compatible with the process, flux/cleaning supplies, stencil and board support, reflow capability and inspection equipment. Quote stencil/panel details after layout is released. Add the PDB-to-Osiris power and I2C harnesses, mating housings/contacts and crimp tooling once connector orientation, wire gauge and length are fixed; they are not included in the 33-item board count.

Hold before production-volume buying

Finalize the Rev B power/load budget and interface, then resolve the provisional 4 A fuse rating, MOSFET fault-energy/SOA and clamp validation. Historical overlaps in INA228 and JST/XT60 connectors are candidates only. Do not stock new Rev B regulators or inductors until their electrical and package requirements are selected.

Decisions for the meeting

Agree a shared-stock budget, nominate someone to inventory Ryan's books, obtain the machine/feeder specification, and choose the initial PDB build quantity. Then request distributor quotes with exact MPN, carrier, pack quantity, lead time, stock and total price. No live prices or stock availability have been quoted in this brief.

Evidence and packaging references

A-P1 source: [procurement/2026-09-13/PDB-grouped-order-list.csv](#); tag `pdb-procurement-2026-09-13`. Exact electrical specifications are retained in the baseline purchasing workbook and datasheets.

Custom reels: <https://forum.digikey.com/t/details-about-digi-reels/1303>

Cut tape packaging: <https://forum.digikey.com/t/a-closer-look-at-taped-packaging-including-cut-tape-and-tape-and-reels/17211>